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QUESTION

“Can **haptic cues** improve pilot recovery from unusual flight attitudes?”

INTERVENTION



We developed a bone-conduction haptic device which activates at unusual attitudes.

The device activates if pitch +25°/-10° or roll ±45°. Based on (Brill et al. 2014), users move “away” from the tapping to achieve straight and level flight. i.e., tapping on the forehead indicates pulling the stick ‘aft’.

INDEPENDENT VARIABLES:

- presence of haptic device
- time of day (level of visual degradation)

DEPENDENT VARIABLES:

- speed of recovery
- workload (NASA TLX)

HARDWARE & SETUP



Participants were tested using Digital Combat Simulator (DCS) simulating a Yak-52. We used a Microsoft SideWinder Joystick, Virpil APC ACE and a Virpil Flight Pedals and Throttle.

RECOMMENDATIONS

- Users found haptics annoying during the day but helpful at night. We recommend having **haptics conditioned on sensory context** (i.e. only providing haptic information when the visual field is ambiguous).
- An **appropriate mental representation can reduce the need for** training. Users who imagined the haptic device as a negative thing (ex. a bee), had better outcomes.

CONDITIONS



- Visually **clear** (day)
- No haptic



- Visually **degraded** (night)
- No haptic



- Visually **clear** (day)
- **Haptic device**



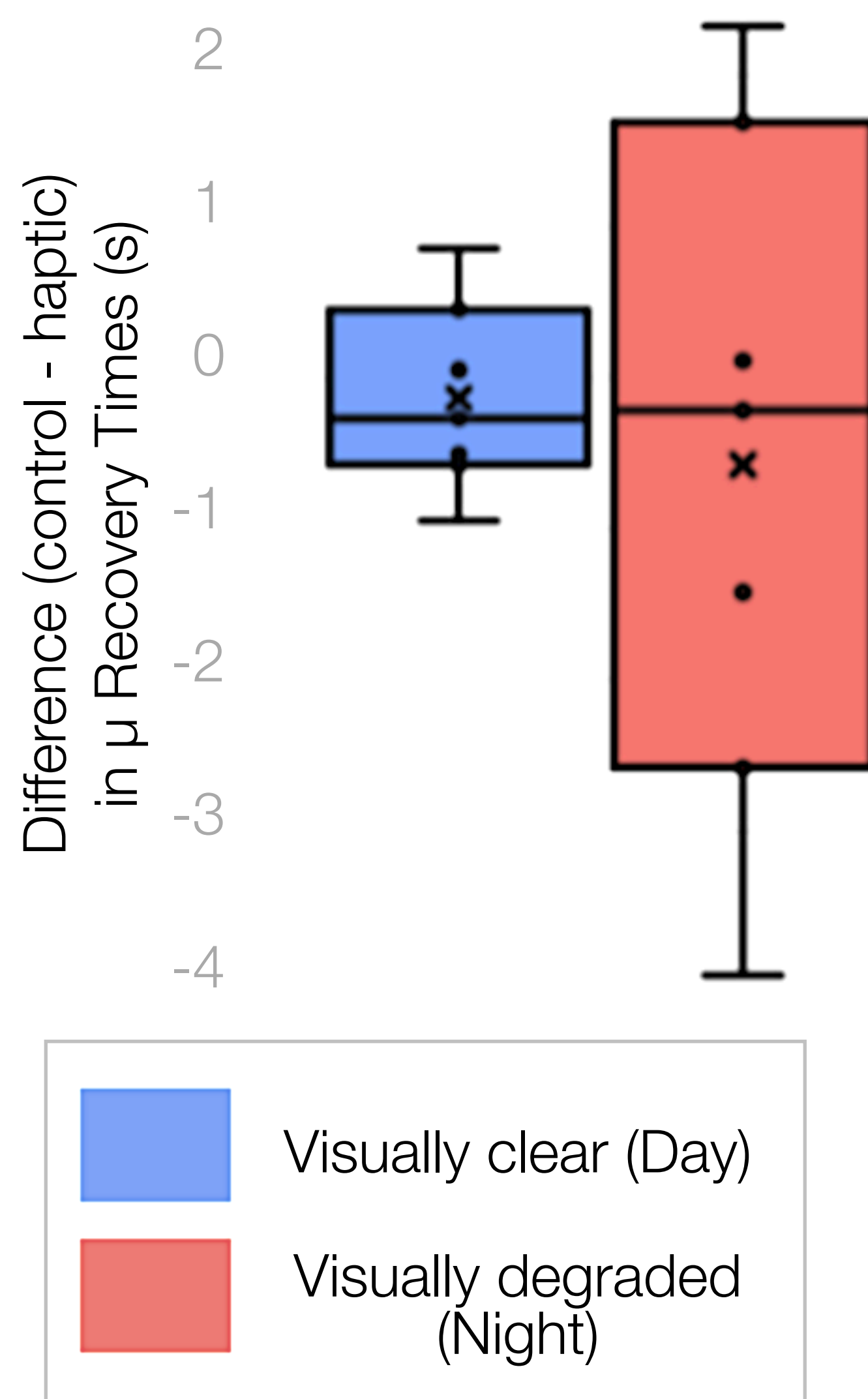
- Visually **degraded** (night)
- **Haptic device**

PROCEDURE

In our experiment, participants were initialized to an unusual attitude and instructed to correct to straight and level flight as quickly as possible.

For each test condition, three unusual flight attitudes were tested for a total of twelve trials per participant. The order of conditions were counterbalanced among participants. Before the task, participants flew freely for six minutes to familiarize with Yak-52 and haptic cues.

RESULTS

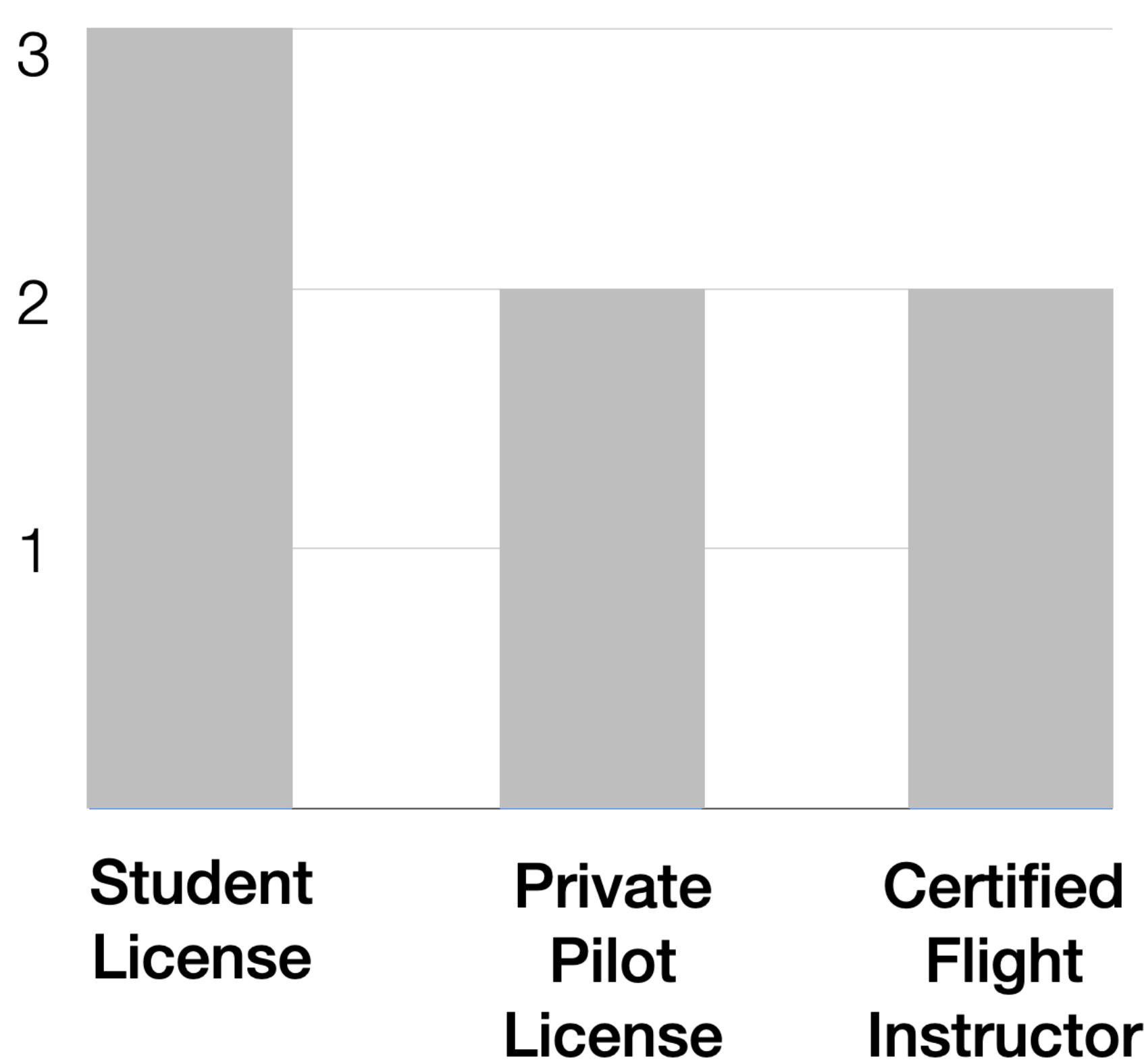


• Effect of adding **haptics varies more** during visual degradation (see graph to the left)

• No significant difference in **recovery times** with haptics ($\mu = -0.36$, $\sigma = 1.44$, $p = 0.892$)

• No significant difference in **workload** with haptics ($\mu = -5.75$, $\sigma = 5.52$, $p = 0.171$).

PARTICIPANTS



n = 7

2 participants were CFIs and 1 had fighter-jet training